ACCRA TECHNICAL UNIVERSITY

FACULTY OF BUSINESS

END OF FIRST SEMESTER EXAMINATION, 2024/2025 ACADEMIC YEAR

COURSE CODE: BAT 203 COURSE TITLE: COST ACCOUNTING 1 (MORNING SESSION)

INSTRUCTION: ANSWER ALL QUESTIONS

TIME ALLOWED: 2.5 HOURS

SECTION A

40 Marks

Answer all questions

Shade the correct answers on the scannable sheet provided

1. When a company uses job order costing, which of the following methods would best allow management to evaluate the efficiency of job-related costs?
   A. Break-even analysis
   B. Variance analysis between applied and actual overhead costs
   C. Depreciation allocation
   D. Activity-based costing (ABC)

   tc = fc +
   y = 400 +10(

2. A company produces 5,000 units of a product at a variable cost of GH₵10 per unit and fixed costs of GH₵40,000. The company wants to reduce its fixed costs by 10% and maintain the same production level. What is the new fixed cost per unit?
   A. GH₵8
   B. GH₵10
   C. GH₵5
   D. GH₵7.20

   vc = 10
   y = 45,000 +10 (5000)
   Cost = 40,000 -4
   10
   100
   50,000 -5
   36000
   4500

3. In a process costing system, how would a company handle abnormal spoilage when calculating the cost of completed units?
   A. Include abnormal spoilage costs in the cost of completed units
   B. Exclude abnormal spoilage costs from both completed and spoilage units
   C. Allocate abnormal spoilage costs to the following period
   D. Adjust the cost of goods sold for abnormal spoilage

4. A company is analyzing its cost structure using CVP analysis. If the contribution margin ratio is 40%, and fixed costs are GH₵120,000, how many units must be sold to achieve a target profit of GH₵30,000, assuming each unit is sold for GH₵25?
   A. 6,000 units
   B. 5,000 units
   C. 7,500 units

D. 9,000 units

5. Under activity-based costing (ABC), what is the primary reason for identifying and assigning costs to cost drivers?
A. To eliminate variable costs from the costing process
B. To calculate break-even points more accurately
C. To simplify the cost allocation process
D. To ensure that indirect costs are allocated based on activities

6. What would be the effect on the break-even point if a company increases its fixed costs but maintains the same variable cost per unit and selling price?
A. The break-even point decreases
B. The break-even point remains unchanged
C. The break-even point increases
D. The break-even point fluctuates based on production levels

7. When using job order costing, how would you treat over-applied overhead at the end of an accounting period?
A. Allocate it to the cost of goods sold
B. Transfer it to finished goods inventory
C. Assign it to work-in-process inventory
D. Ignore it as it has no impact on net income

8. A company is considering using standard costing for overhead allocation. How would an increase in the actual production volume, without an increase in fixed costs, affect the overhead allocation rate?
A. The allocation rate increases
B. The allocation rate remains unchanged
C. The allocation rate decreases
D. The allocation rate fluctuates depending on the level of sales

9. Under the FIFO method in process costing, how are costs assigned to the units in the beginning inventory?
A. Costs are allocated based on the percentage of completion at the start of the period
B. Costs are assigned using a weighted average method
C. Costs from the previous period are excluded from the calculation
D. Costs are assigned evenly across all units

10. Which of the following cost behavior patterns would most likely result from economies of scale as production levels increase?
A. Fixed costs remain constant, while variable costs increase
B. Fixed costs decrease as production increases
C. Variable costs increase at a decreasing rate
D. Total costs remain constant regardless of production changes

11. In cost-volume-profit (CVP) analysis, the margin of safety measures
A. The difference between actual sales and break-even sales
B. The fixed cost coverage ratio
C. The amount by which variable costs exceed fixed costs
D. The degree of operating leverage in the company

12. A company has a contribution margin of 30%, fixed costs of GH₵80,000, and sales of GH₵300,000. What is the company's margin of safety in units, assuming the selling price per unit is GH₵50?
A. 5,000 units
B. 2,000 units
C. 3,000 units
D. 4,000 units

13. In job order costing, if the company applies overhead based on machine hours, and actual machine hours are higher than the estimated hours, the company will
A. Have over-applied overhead
B. Have under-applied overhead
C. Correct the overhead rate
D. Reduce fixed costs

14. Which of the following best describes the role of cost allocation in pricing decisions?
A. Allocating all indirect costs to increase profitability
B. Assigning both direct and indirect costs to reflect the true cost of production
C. Only applying variable costs to determine product prices
D. Using overhead costs exclusively to establish product prices

15. A company is using process costing and reports that normal spoilage is expected to be 5% of units started. How would the cost of normal spoilage be treated in the costing system?
A. Included in the cost of completed units
B. Excluded from both the cost of goods sold and completed units
C. Transferred to finished goods inventory
D. Absorbed by the next period's production costs

16. If the actual overhead costs are greater than the applied overhead costs in job order costing, what is the effect on the income statement?
A. Over-applied overhead increases income
B. Under-applied overhead decreases income
C. Over-applied overhead reduces income
D. Under-applied overhead has no effect on income

17. In process costing, when are costs transferred from one department to the next?
A. At the end of each accounting period
B. When the products are completed in the first department
C. As soon as products are started in the department
D. When products are sold to customers

18. Which of the following is an advantage of using activity-based costing (ABC) over traditional costing methods?

25. The primary difference between job order costing and process costing is
A. Process costing accumulates costs for individual units
B. Job order costing tracks costs for each department
C. Process costing is used for unique products, and job order costing is for mass-produced goods
D. Process costing accumulates costs by process, while job order costing accumulates costs by job

26. In activity-based costing, what is the impact of a more accurate identification of cost drivers on product costing?
A. Products will be allocated a higher proportion of indirect costs
B. Products will be allocated a lower proportion of direct costs
C. The cost allocation will be more precise and reflect actual resource usage
D. The overall cost of the products will decrease

27. Which of the following is a limitation of cost-volume-profit (CVP) analysis?
A. Assumes all costs are variable
B. Ignores the impact of fixed costs
C. Assumes constant sales prices and costs
D. Only applies to companies with low operating leverage

28. How does an increase in sales volume affect operating leverage?
A. Increases operating leverage
B. Decreases operating leverage
C. Has no impact on operating leverage
D. Depends on the fixed cost structure

29. The break-even point for a company selling a product at GH₵15 with variable costs of GH₵5 per unit and fixed costs of GH₵50,000 is
A. 5,000 units
B. 3,333 units
C. 10,000 units
D. 1,500 units

30. Under job order costing, how do you calculate the cost of goods manufactured at the end of a period?
A. Add direct materials used and direct labor to applied overhead
B. Add the total of work-in-process to the finished goods inventory
C. Deduct beginning finished goods from total costs incurred
D. Add direct material costs to total overhead costs

. A company decides to use a higher markup percentage on products with higher variable costs. What is the likely outcome of this pricing strategy?
A. Higher variable costs will increase the overall contribution margin
B. The fixed costs will be spread over a larger volume of sales
C. The higher markup percentage will reduce the need for cost analysis

A. Simplicity in cost allocation
B. More accurate cost assignment to products based on activities
C. Reduced cost of overhead allocation
D. Focuses only on variable costs

19. How would an increase in variable costs per unit affect the break-even point if fixed costs remain unchanged?
A. Increase the break-even point
B. Decrease the break-even point
C. Leave the break-even point unchanged
D. Cause a fluctuation in the break-even point

20. Which method of overhead allocation is most appropriate for a company with multiple cost drivers and complex production processes?
A. Traditional costing
B. Standard costing
C. Activity-based costing (ABC)
D. Direct costing

21. In CVP analysis, what is the impact of a change in the contribution margin ratio on the required sales volume to break even?
A. Higher contribution margin ratio decreases the break-even sales volume
B. Higher contribution margin ratio increases the break-even sales volume
C. The break-even sales volume remains unchanged
D. The break-even sales volume increases linearly with variable costs

22. A company has total fixed costs of GH₵50,000, and each unit is sold for GH₵40. If the variable cost per unit is GH₵25, what is the break-even point in units?
A. 2,000 units
B. 1,000 units
C. 500 units
D. 3,000 units

23. If a company's overhead allocation rate is based on direct labor hours, but labor costs decrease significantly, what is the likely impact on the allocation of overhead costs?
A. Overhead allocation per unit will increase
B. Overhead allocation per unit will decrease
C. Overhead allocation per unit will remain the same
D. The allocation base will become irrelevant

24. What is the effect of using a predetermined overhead rate versus actual overhead costs in job order costing?
A. Predetermined overhead rates provide a more accurate reflection of actual costs
B. Predetermined overhead rates help smooth out fluctuations in overhead costs
C. Actual overhead costs are always preferred in job order costing
D. Predetermined rates reduce the need for cost control

For Skyline Corporation, a customized delivery fee of GH₵900 was necessary to meet the client's specifications.

Overheads:

Indirect Material Costs: GH₵3,000, allocated based on direct material costs.
Indirect Labor Costs: GH₵2,400, allocated based on direct labor costs.
Indirect Expenses: GH₵1,500, allocated equally across all three orders.

Pricing Policy:

Royal Woodcraft Ltd. applies a 35% mark-up on the total cost of production to determine the selling price of each order.

Required (Total: 10 marks):

(a) Calculate the Prime Cost for each order. *(3 marks)*
(b) Calculate the Total Cost of Production for each order. *(3 marks)*
(c) Calculate the Selling Price for each order based on a 35% mark-up on the total cost of production. *(1.5 marks)*
(d) Discuss one advantage of using a mark-up on the total cost of production for determining selling prices in custom furniture manufacturing. *(1.5 marks)*

Question 2.

OakCraft Ltd. is a custom cabinet manufacturing company that received an order in October 2024 to create 30 high-quality, customized kitchen cabinets for a luxury apartment project. The order, labeled as Job No. 202, has specific cost details related to direct materials, direct labor, production overheads, non-production overheads, and sales information.

The direct materials cost for Job No. 202 included 150 cubic feet of solid oak wood at GH₵90 per cubic foot, totaling GH₵13,500. In addition, the stain and finish materials amounted to GH₵4,200. Regarding direct labor costs, OakCraft Ltd. employed cabinet makers and finishers. Cabinet makers worked 300 hours at a rate of GH₵25 per hour, resulting in a total direct labor cost of GH₵7,500. Finishers worked 100 hours at a rate of GH₵30 per hour, totaling GH₵3,000.

Production overheads for Job No. 202 are allocated based on direct labor hours, using a predetermined overhead absorption rate of GH₵18 per labor hour. The total production overheads for Job No. 202 are calculated by multiplying the total labor hours (300 + 100 = 400 hours) by the absorption rate of GH₵18, resulting in GH₵7,200.

Non-production overheads, which include general and administrative expenses, amounted to GH₵4,800 for October. Since OakCraft Ltd. completed three jobs in October, these overheads are allocated based on the number of jobs completed. Therefore, Job No. 202 is allocated GH₵1,600 (GH₵4,800 ÷ 3 jobs).

The client agreed to pay GH₵85,000 for the 30 cabinets.

Required:

(a) Prepare a Job Cost Sheet for Job No. 202, showing all cost details for direct materials, direct labor, and overheads. (4 marks)
(b) Compute the Total Cost of Job No. 202. (2 marks)
(c) Determine the Profit or Loss on Job No. 202, given the client's payment. (3 marks)
(d) Discuss the importance of overhead absorption in job order costing and its impact on profitability. (1 mark)

Question 3

The following data relates to the production activity and total costs of ABC Ltd. for the first six months of 2024:

Month	Activity (units)	Total Cost (GH₵)
January	100	1,200
February	200	1,800
March	150	1,500
April	250	2,100
May	300	2,500
June	350	3,000
	400	5850
	450	9120
	300	12720
	150	16620
	600	

Required:

Using regression analysis:

(a) Determine the cost function in the form Cost (GH₵) = $a + bx$, where a is the fixed cost and is the variable cost per unit. (5 marks)
(b) Interpret the regression coefficients a and b. (2 marks)
(c) Predict the total cost if the activity level is 400 units and 1000 units. (3 marks)

D. The company may experience lower sales volume

32. In a cost accounting system, which cost classification method would be most effective for controlling product costs in a high-volume production environment?
A. Standard costing
B. Activity-based costing
C. Absorption costing
D. Job order costing

33. How would an increase in production volume affect the unit cost under absorption costing?
A. Unit cost would remain the same regardless of production volume
B. Unit cost would decrease as fixed costs are spread over more units
C. Unit cost would increase as production volume increases
D. Only direct labor costs would decrease per unit

34. The allocation of overhead costs in a process costing system is based primarily on
A. Direct material consumption
B. Direct labor hours used
C. Machine hours or another activity base
D. Total production output

35. What is the most appropriate method for assigning overhead costs to individual products in a manufacturing process with significant variations in production complexity?
A. Absorption costing
B. Job order costing
C. Activity-based costing (ABC)
D. Standard costing

36. In process costing, the treatment of joint costs for products with different selling prices is typically based on
A. The number of units produced
B. The amount of direct labor used
C. The direct material costs of each product
D. The relative sales value of each product

7. Which of the following cost drivers is most likely to be relevant in a highly automated production environment?
A. Machine hours
B. Direct labor hours
C. Number of units produced
D. Number of workers involved

In job order costing, if overhead is under-applied, the company must
A. Increase the overhead rate for the next period
B. Charge the under-applied amount to work-in-process inventory
C. Reduce fixed costs for the next period
D. Allocate the under-applied overhead to finished goods inventory

39. The primary purpose of variance analysis in cost accounting is to
A. Determine the best production method
B. Identify discrepancies between expected and actual costs
C. Reduce the overall cost of production
D. Allocate indirect costs to products

40. Which of the following methods is most likely to provide an accurate reflection of costs in a production process with multiple cost drivers?
A. Activity-based costing (ABC)
B. Traditional costing
C. Standard costing
D. Job order costing

Section B

Answer all questions in the answer booklet provided

Question 1

Royal Woodcraft Ltd. specializes in crafting premium, custom-made conference tables for corporate clients. In October 2024, the company completed three major orders for different clients: Greenfield Inc., Prestige Realty, and Skyline Corporation. Below is the information on production costs for each order:

Direct Materials Cost (Oak Wood):
Greenfield Inc.: GH₵9,800
Prestige Realty: GH₵12,500
Skyline Corporation: GH₵14,300

Direct Labor Costs:
Royal Woodcraft Ltd. employs two types of skilled workers: woodworkers and finishers.
Woodworkers worked a total of 160 hours at a rate of GH₵28 per hour.
Finishers worked a total of 120 hours at a rate of GH₵22 per hour.
Labor hours were allocated to each order as follows:

Client	Woodworker Hours	Finisher Hours
Greenfield Inc.	50	40
Prestige Realty	70	50
Skyline Corporation	40	30

Direct Expenses:
For Greenfield Inc., Royal Woodcraft Ltd. paid a certification fee of GH₵1,000 for premium-grade wood.
For Prestige Realty, special packaging charges of GH₵650 were incurred for international shipping requirements.